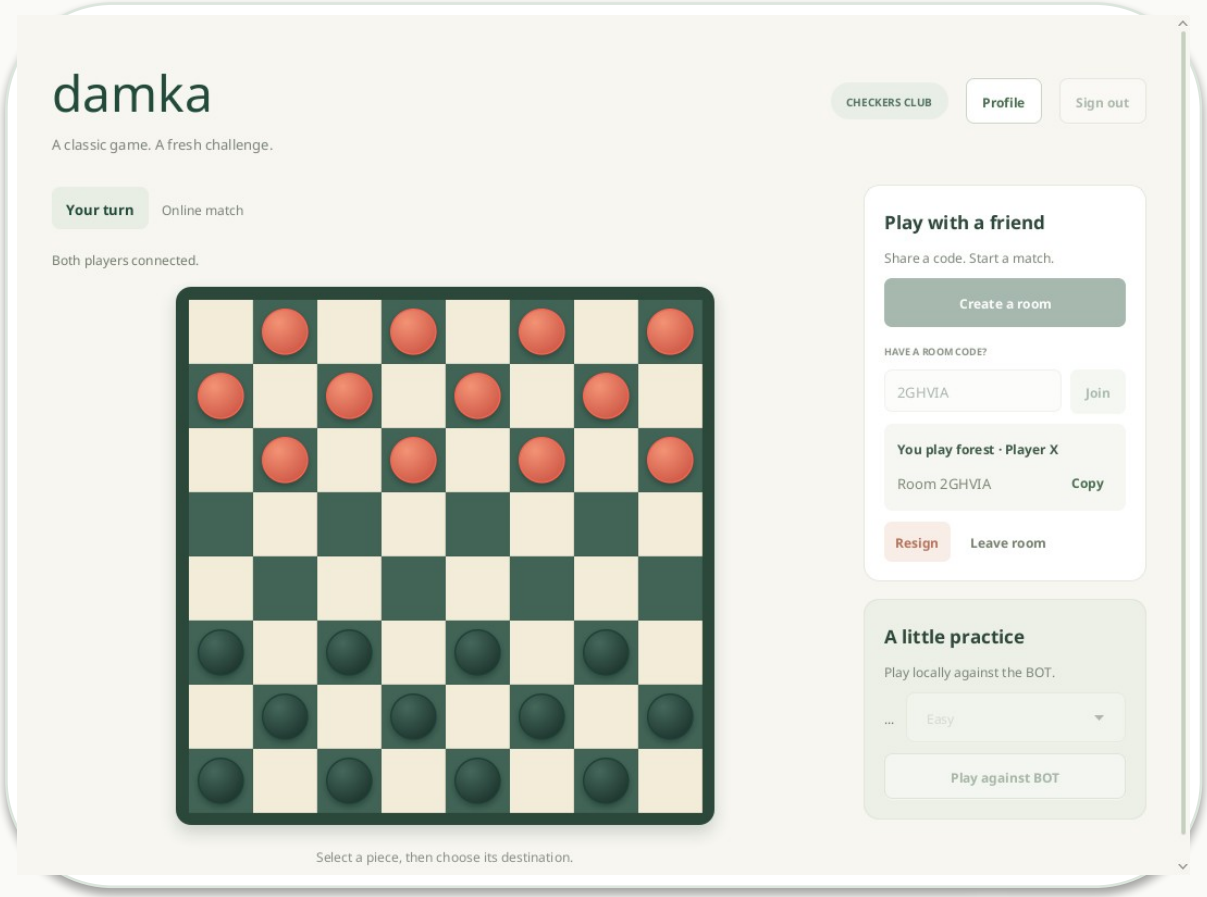
Damka

Network checkers.
A shared game,
a lasting profile.

JavaFX interface
TCP multiplayer · Local BOT
SQLite accounts and history

Muthanna Farhat · Rabie Abu Aad

Supervisor: Inbal Ronen



One game, two ways to play

Keep both screens in sync — and remember the result.

Online matches

Two computers
One room code

Local BOT

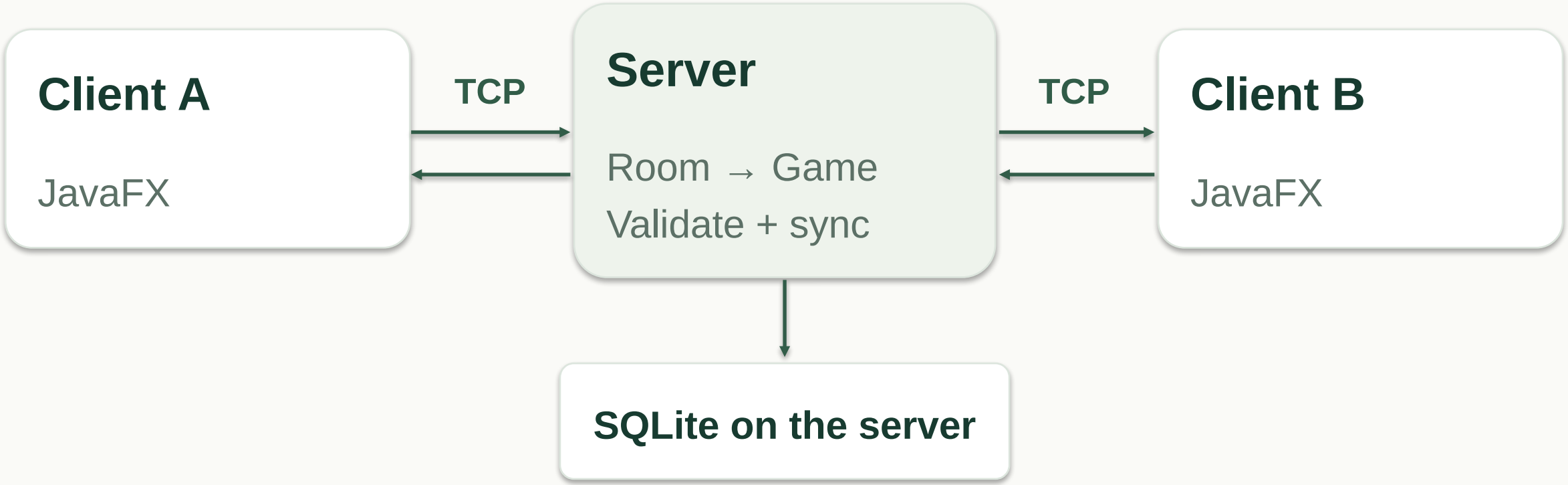
Three difficulty levels
Offline practice

Personal profiles

Accounts and statistics
Persistent match history

One shared rules engine: Game

Why a client–server design?



Choice: persistent TCP for two-way updates

Alternative: HTTP polling

From a click to both screens

Client A

Select a piece and destination

```
MOVE 5 0 4 1
```

Server

Room: identity + turn

Game.move(): rules

Accept or reject

Both clients

Accepted move

```
1   MOVE_APPLIED
2   BOARD
3   TURN
```

Animate → display the accepted state

UTF-8 · one message per line · network work off the UI thread

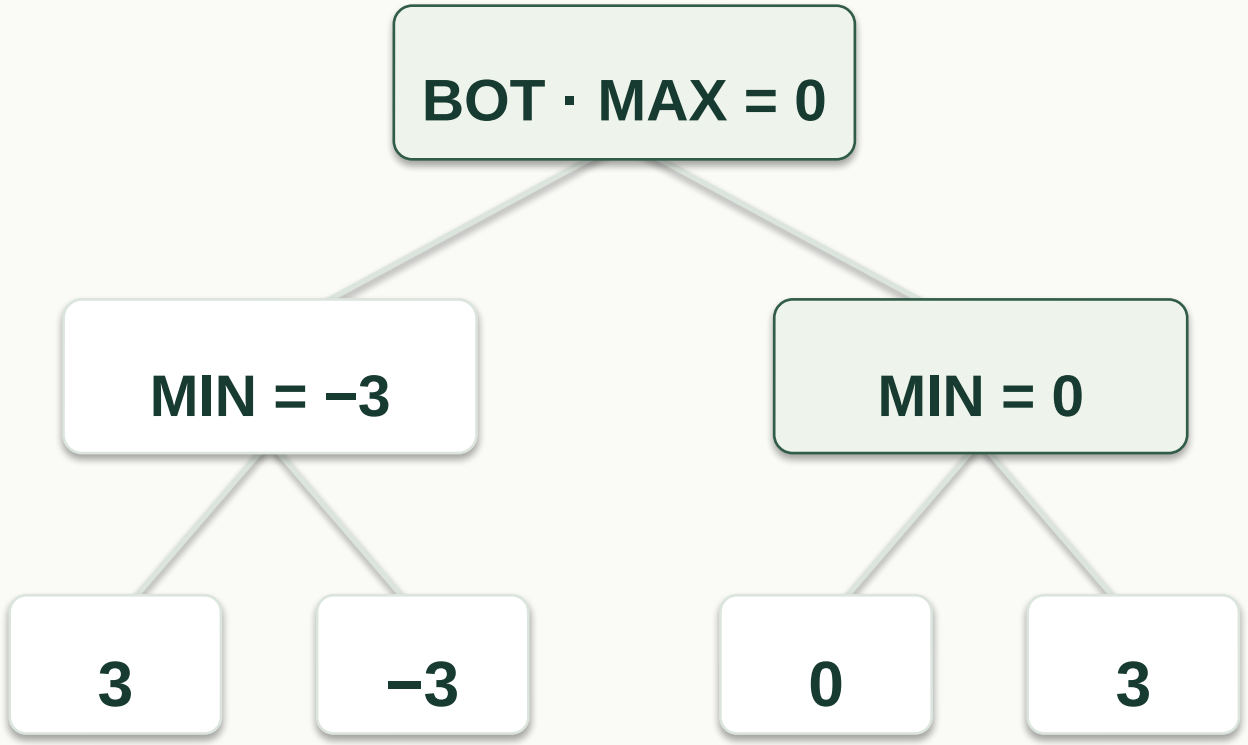
Minimax: how the BOT chooses

One rules engine: Game

Legal moves on copied states

Man = 3 King = 6

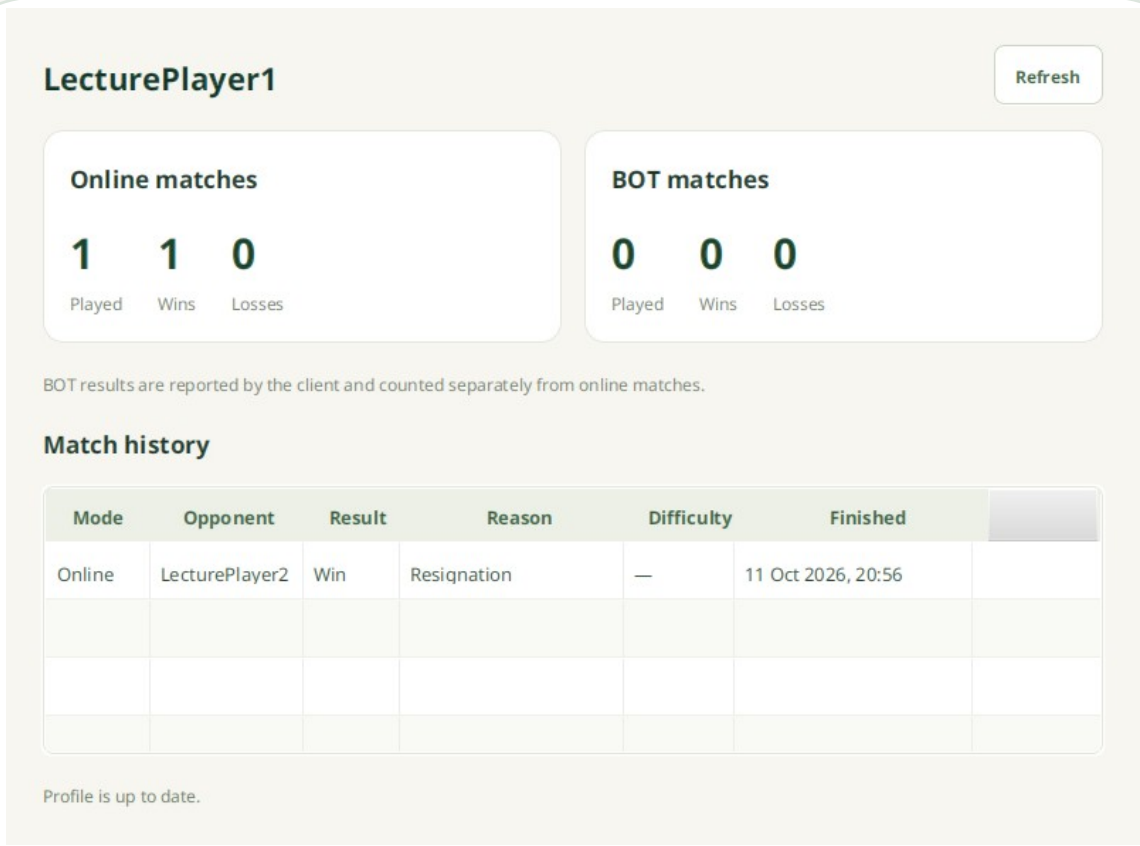
Search depth: 1 / 2 / 3



Illustrative two-step search

Alpha–Beta skips branches that cannot improve the decision.

Accounts that remember results



Demo account · sample history

`profiles`

Username · salt · password hash

`game_results`

Players · winner · mode · time

Statistics are derived from results.

Verification and AI-assisted development

48 **automated tests**

Game rules and edge cases
Real TCP protocol checks
SQLite persistence and deduplication

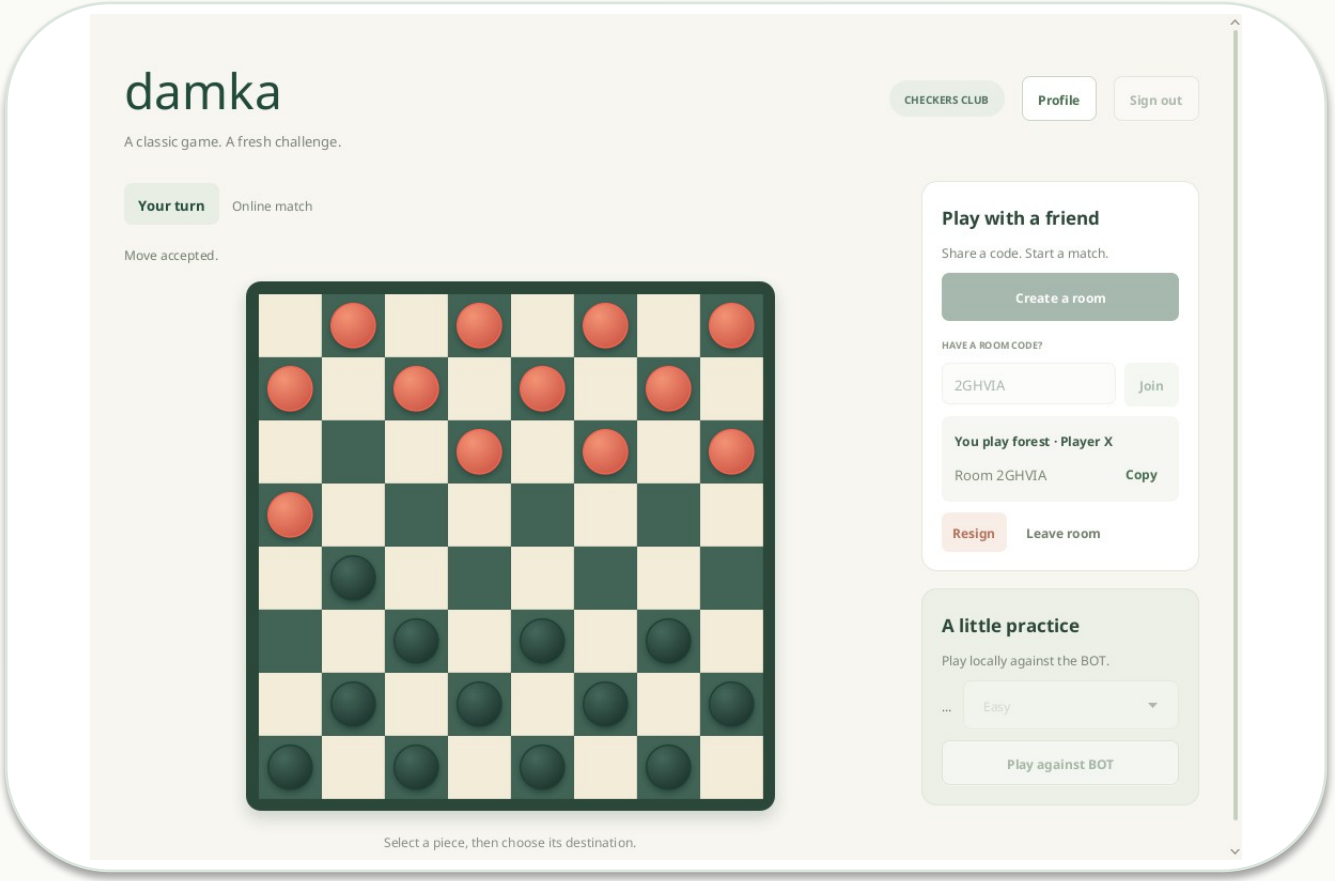
Documented passing run: 12 Oct 2026

Codex + GitHub Copilot

Planning · code · explanations
Tests and documentation

Read → run → review → refine

Live demo — two players, one match



1 **Log in**

Two different accounts

2 **Create / join**

Share one room code

3 **Move / synchronize**

Both clients show the update

4 **Resign / profile**

One result, saved once

Current limits and next step

LAN focus

No TLS in this version

No match resume

A detected disconnect ends the active match

Local BOT results

Client-reported

No offline upload queue

NEXT **In-room chat over the existing TCP connection**

What we built and learned

One authoritative game.
Two synchronized clients.
Persistent player results.

Questions?

Game rules · Networking
BOT · Data storage

Rabie Abu Aad — Game Logic
Muthanna Farhat — TCP Communication

github.com/MuthannaFrhat/damka-game